

PRASANNA REDDY PULAKURTHI

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EDUCATION

Rochester Institute of Technology

Doctor of Philosophy (Ph.D.) – Electrical and Computer Engineering | GPA: 3.93/4.0

Rochester, NY
Aug 2019 to Dec 2025 (expected)

Rochester Institute of Technology

Master of Science (MS) – Electrical Engineering | GPA: 4.0/4.0

Rochester, NY
Aug 2017 to Jul 2019

PESIT Bangalore South Campus

Bachelor of Engineering (BE) – Electronics and Communications Engineering

Bangalore, India
Aug 2013 to Jun 2017

TECHNICAL SKILLS

- **LANGUAGES & SOFTWARE TOOLS:** Python, C, HTML/CSS, MATLAB, Linux, Unix, ROS, Gazebo, Docker, LaTeX, Git, SQL, Blender
- **AI/ML:** PyTorch, TensorFlow, TensorBoard, Keras, CUDA, GPU, PyTorch DDP, TensorRT, ONNX, NumPy, Scikit-Learn, data structures, OpenCV, mmCV, Hugging Face, Transformers, LLM, NLP, MMAction2, Path Planning, ORB-SLAM, CLIP, OpenAI, BLIP-2, LLaVA, Qwen-VL, MLLM

PROFESSIONAL EXPERIENCE

Ph.D. Researcher – Machine Learning & Computer Vision

Rochester Institute of Technology

Aug 2019 to Present

- **Multimodal Reasoning (X-CoT, EMNLP 2025):** Designed an LLM-based chain-of-thought retrieval system improving R@1 by +5.6% on MSVD and producing explainable text-video retrieval.
- Developed two novel **Data Augmentation** techniques (1. Dual-region and 2. Shuffle PatchMix) to improve model robustness and generalization. Demonstrated improvements in performance on domain adaptation datasets such as PACS by up to **+7.3 pp** (percentage points), and person re-identification datasets such as Market-1501 by up to **+6.91 pp**.

Perception Engineer Intern – DEVCOM Army Research Laboratory

via Booz Allen Hamilton and MBO Partners

May 2023 to Aug 2023

- **Advanced Generative Models:** Developed a GAN variant combining neural-architecture search, PMish activation, bounded MMD repulsive loss, and adaptive rank decomposition, achieving a reduction of FID scores by **34%** (CIFAR-10) and **28%** (STL-10) while compressing the generator **4–10×**, enabling real-time edge deployment.
- **Data Generation:** Generated new synthetic gesture video data for Human Action Recognition using Generative Adversarial Network and significantly lifted human-action recognition accuracy across ground (**+4.4 pp**) & aerial (**+7.7 pp**) views.

Perception Engineer Intern – DEVCOM Army Research Laboratory

via Booz Allen Hamilton and MBO Partners

May 2022 to Aug 2022

- Developed domain adaptation techniques to improve model robustness across different digit domains, lifting cross-domain accuracy by up to **+2.56 pp** on difficult shifts.
- **Globally-optimal Iterative ML Classifier:** Formulated a fast, globally-optimal Iterative Maximum Likelihood Classifier by deriving a fixed-point solver from a regularized ML objective, achieving $\sim 3\times$ faster convergence compared to gradient descent.

Engineering Intern – Qualcomm (VR Team)

May 2021 to Aug 2021

- Developed a novel temperature-compensation algorithm enhancing the accuracy of the 6DOF 4Tx ultrasound VR controller.
- Implemented the algorithm in C, improving the root mean squared error (RMSE) by **+11.29 pp** on synthetic log data.

Engineering Intern – Microsoft (Iris Recognition, HoloLens-2 AR Team)

May 2020 to Aug 2020

- Optimized Gabor filter frequencies in Iris Recognition pipeline, significantly enhancing signature uniqueness.
- Developed GAN-based Super Resolution to enhance low-resolution Iris image quality while retaining key Gabor frequencies.

Research Assistant – Oak Ridge National Laboratory [[Link](#)]

May 2018 to Jul 2019

- Designed and executed an unmanned aerial system (UAS) data collection to capture aerial imagery of a building casting shadows on 14 differently colored panels, both in and out of shadow.
- Developed six algorithms for shadow detection in aerial images using color transforms and machine learning techniques, leveraging collected data. Improved upon traditional approaches by **+1.51%** in accuracy and **+2.4%** in F-score.

SELECT PUBLICATIONS

- **Prasanna Reddy Pulakurthi** et al. "X-CoT: Explainable Text-to-Video Retrieval via Large Language Models (LLM) based Chain-of-Thought Reasoning," *EMNLP*, 2025. [[Link](#)]
- **Prasanna Reddy Pulakurthi** et al. "Automatic Instruction Programming for Multimodal Large Language Models (MLLM) Based Text-Video Retrieval," submit to *CVPR* 2026.
- **Prasanna Reddy Pulakurthi** et al. "Shuffle PatchMix Augmentation with Confidence Margin Weighted Pseudo Labels for Enhanced Source-Free Domain Adaptation," *IEEE ICIP*, 2025. [[Link](#)]
- **Prasanna Reddy Pulakurthi** et al. "Effective dual-region augmentation for reduced reliance on large amounts of labeled data," *Synthetic Data for Artificial Intelligence and Machine Learning: Tools, Techniques, and Applications III*. SPIE, 2025. [[Link](#)]
- **Prasanna Reddy Pulakurthi** et al. "Enhancing GANs with MMD Neural Architecture Search, PMish Activation Function and Adaptive Rank Decomposition," *IEEE Access*, 2024. [[Link](#)]
- **Prasanna Reddy Pulakurthi** et al. "Enhancing human action recognition with GAN-based data augmentation," *Synthetic Data for Artificial Intelligence and Machine Learning: Tools, Techniques, and Applications II*. SPIE, 2024. [[Link](#)]
- **Prasanna Reddy Pulakurthi** et al. "Enhancing GAN Performance through Neural Architecture Search and Tensor Decomposition," *IEEE ICASSP*, 2024. [[Link](#)]

PROJECTS

Machine Learning and Generative AI Projects

- Utilized GANs to generate a synthetic video dataset for Human Action Recognition. [[Link](#)]
- Applied Generative Adversarial Networks (GANs) to unsupervised/source-free domain adaptation and image-to-image translation tasks. [[Link](#)]
- Employed Stable Diffusion with Control-Net to create a gesture recognition dataset with multiple views.
- Developed Invertible Neural Networks (INNs) to perform super-resolution and image denoising. [[Link](#)]

Vision Captioning Assistant (V-CAT) [[Link](#)]

- Built a full-stack, camera-driven "smart-glasses" assistant that streams live video to vision-language models and returns real-time scene captions plus synthesized speech for hands-free accessibility.

Signmentor AI (ASL Sign Tutoring) [[Link](#)]

- Built a full-stack website for ASL sign tutor that accurately identifies every handshape in the ASL alphabet (A to Z) by pairing your camera feed with cutting-edge Multi-modal LLM for fast, reliable classification and instant learner feedback.

Comparison of Simultaneous Localization and Mapping (SLAM) Algorithms in ROS Environment [[Link](#)]

- Simulated and benchmarked Gmapping (2D laser-based), Octomap (probabilistic 3-D occupancy-grid), and RTAB-Map (RGB-D dense mapping) in a Gazebo/ROS indoor environment using custom and stock robots outfitted with LiDAR and RGB-D sensors.
- Evaluated map quality, runtime, and sensor cost, recommending Gmapping for lightweight real-time 2-D mapping, Octomap for memory-efficient 3-D obstacle maps, and RTAB-Map for high-fidelity textured reconstructions.

Image and Signal Processing Projects

- Implemented several image and signal processing algorithms, such as image resizing, noise filtering, image enhancement, image segmentation, camera calibration, stereo reconstruction, affine, and projective transforms.

Real-Time Attendance System using Image Processing [[Link](#)]

- Developed a real-time automatic attendance system using OpenCV for face detection, Python-implemented Binarized Statistical Image Features for face recognition, and a Kivy-based app interfacing via sockets and PHP with an Apache-hosted GUI displaying attendance data stored in MySQL.

Automatic License Plate Detection and Recognition [[Link](#)]

- Built an automatic number plate detection and recognition system using edge and histogram-based image processing methods for detection and neural networks for recognition, coded in MATLAB.

COURSEWORK

Digital Signal Processing, Digital Image Processing, Multi-view Imaging, Artificial Intelligence Explorations, Pattern Recognition, Information Theory Coding, Image and Video Compression, Advanced Engineering Mathematics